

AUTISM SPECTRUM DISORDER ANALYSIS FOR TODDLERS

Batch No : B20

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Abstract

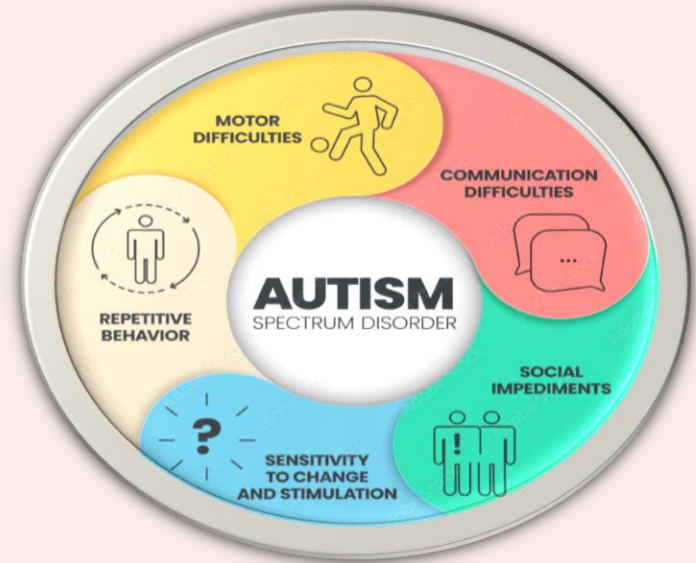
- ❖ Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that impacts communication, social interaction, and behavior.
- ❖ Early detection is essential for improving outcomes through timely interventions.
- ❖ This project presents a web-based autism screening tool that uses responses to 10 key behavioral questions to assess ASD risk levels.
- ❖ A trained machine learning (ML) model classifies results into "Mild," "Severe," or "No ASD Traits" and provides relevant precautionary recommendations.

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Introduction to Autism Detection

- ❖ This system begins by collecting basic information about the child, such as age and gender, followed by 10 carefully selected behavioral questions.
- ❖ Based on the responses, a trained machine learning model evaluates the risk of ASD and classifies it as "No ASD Traits," "Mild," or "Severe."
- ❖ The system then provides precautionary recommendations and possible treatment steps, offering parents and caregivers an accessible and effective tool for early autism screening.



Problem statement

Develop a machine learning-based system for early detection of ASD and recommendation of personalized treatment plans.

- ASD is a complex neurodevelopmental disorder affecting social interaction, communication, and behavior.
- Early intervention significantly improves long-term outcomes.
- Traditional diagnosis is often subjective, time-consuming, and reliant on specialist availability.

Literature survey

Author(s)	Tool(s) / Dataset(s)	Contribution / Method
Wall et al. (2012)	ADI-R, SVM	Reduced 93 ADI-R items to 7; achieved 99% accuracy using SVM.
Duda et al. (2016)	ADOS, ML classifiers	Used ADOS data for classification; achieved 96.8% accuracy.
Heinsfeld et al. (2018)	ABIDE (fMRI), Deep Learning (CNN)	Brain imaging-based detection; ~70% accuracy.
Hashemi et al. (2015)	3D facial imaging	Used facial morphology of toddlers to detect ASD.
Khalifa et al. (2021)	Speech, Eye-tracking data, ANN/CNN	Achieved >90% accuracy using deep learning.
UCI ASD Dataset	Behavioral questionnaire (M-CHAT, Q-CHAT)	Used in various ML models for severity classification.

Literature survey (Cont...)

- ❖ Autism detection is rapidly evolving with the help of AI and ML.
- ❖ While traditional methods remain the gold standard, automated tools offer scalable and accessible alternatives.
- ❖ Ongoing research aims to enhance the accuracy, interpretability, and accessibility of ASD detection tools to enable early intervention and better outcomes.

Existing Systems

Autism detection has traditionally relied on standardized clinical tools such as the **Autism Diagnostic Observation Schedule (ADOS)** and the **Autism Diagnostic Interview – Revised (ADI-R)**.

These tools are comprehensive and widely accepted but require trained professionals and are time-consuming.

Disadvantages

Limited accessibility: Requires expert evaluation, which may not be available in remote areas.

Subjectivity: Depends on parental responses and clinical judgment, leading to inconsistencies.

Lack of instant feedback: Parents must wait for expert assessment and guidance.

Proposed system

- ❖ A web-based application that provides an automated autism screening test for toddlers.
- ❖ Uses a machine learning (ML) model to analyze responses from a 10-question test.
- ❖ Provides immediate classification results and treatment recommendations.

Advantages

- ❖ **Faster screening:** Instant analysis and severity classification.
- ❖ **Accessible:** Can be used by parents and caregivers from any location.
- ❖ **Objective results:** Uses ML for data-driven predictions.
- ❖ **Guidance & support:** Offers precautionary measures and possible treatment options.

Functional Requirements

- ❖ **Welcome Interface:** The system begins with a welcoming interface that serves as the landing page. This page introduces the purpose of the platform
- ❖ **Basic Information Input Form:** Once the user initiates the screening, the system presents a structured form to collect basic demographic and health-related information about the toddler.
- ❖ **Questionnaire Module:** After the basic information is submitted, users are guided through the Q-Chat-10 questionnaire.
- ❖ **ML model-based risk prediction:** The model receives the input and tells the risks based on the user input

Functional Requirements

- ❖ **Input Validation**
- ❖ **Risk Prediction Engine**
- ❖ **Result Summary**
- ❖ **Display Recommendation Generator**
- ❖ **Admin and Feedback Module**

Non-Functional Requirements

- ❖ **Usability:** The application is designed to be intuitive and user-friendly. Special care is taken to make it accessible for individuals with little to no technical background.
- ❖ **Responsiveness:** All user actions, such as navigation between questions or receiving prediction results, occur within a few seconds.
- ❖ **Security:** Security is paramount, especially because the application handles sensitive information.
- ❖ **Cross-Platform Compatibility:** The application is web-based and tested to function seamlessly across common browsers like Chrome, Firefox, and Edge.

Non-Functional Requirements

- ❖ **Scalability and maintainability**
- ❖ **Reliability & Availability**
- ❖ **Maintainability**
- ❖ **Accessibility Compliance**

HardWare requirements

- ❖ **Processor:** Intel Core i3
- ❖ **RAM:** 4 GB minimum
- ❖ **Hard Disk:** 40 GB
- ❖ **Internet Connection:** Required for accessing web modules and downloading dependencies

Software requirements

- ❖ **Operating System:** Windows 7
- ❖ **Web Browser:** Google Chrome
- ❖ **Software Development Tools:** Visual Studio Code, Jupyter Notebook
- ❖ **Programming Language:** Python 3.7
- ❖ **Web Framework:** Flask
- ❖ **Packages:** NumPy, Pandas, Scikit-learn, XGBoost, Joblib, Jinja2
- ❖ **ML Toolkit:** scikit-learn and XGBoost

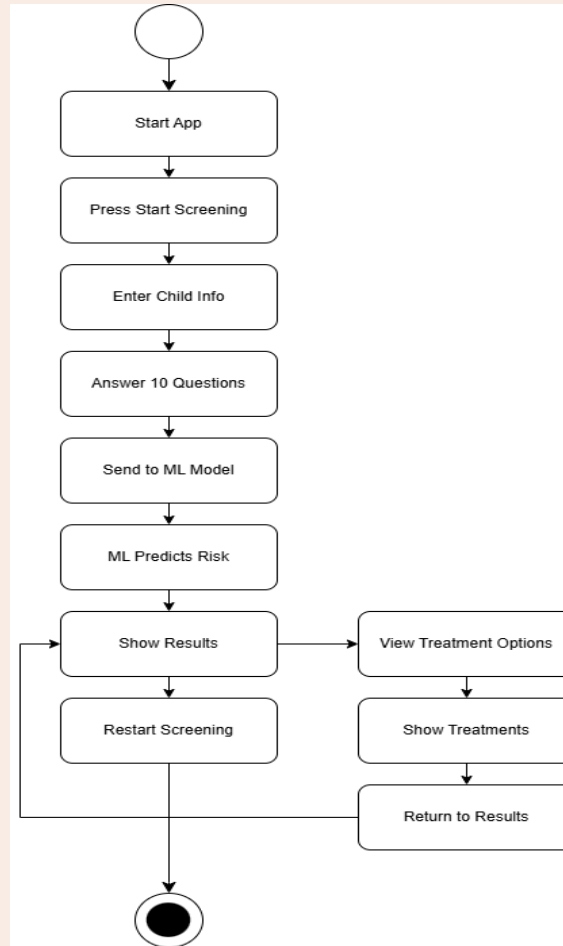
System Architecture Overview

- ❖ **Web-based** screening system with frontend, backend, and ML model
- ❖ Frontend **collects** user data and displays results
- ❖ Backend handles **logic, validation, and model inference**
- ❖ ML model predicts **ASD risk** from user input

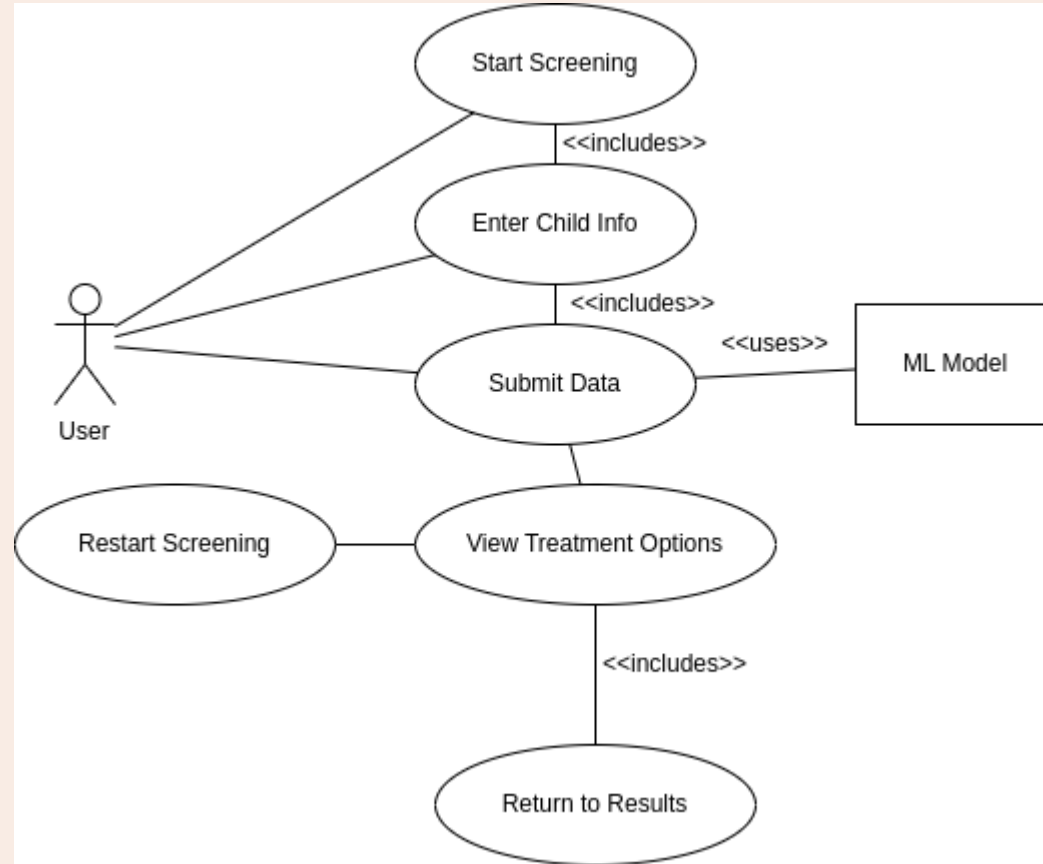
Modules

- ❖ **Authentication Module:** Manages user accounts, session control, and ensures secure access to sensitive screening data.
- ❖ **Data Collection Module:** Captures child-specific inputs and standardized questionnaire responses through a user-friendly interface.
- ❖ **Preprocessing Module:** Converts raw user input into clean, structured feature vectors suitable for machine learning inference.
- ❖ **Prediction Module:** Applies a trained ML model to generate ASD risk scores categorized into Low, Medium, or High.
- ❖ **Result Module:** Presents an understandable risk assessment along with guidance for follow-up action.
- ❖ **Reporting Module:** Generates downloadable reports with screening results, timestamps, and suggested next steps.

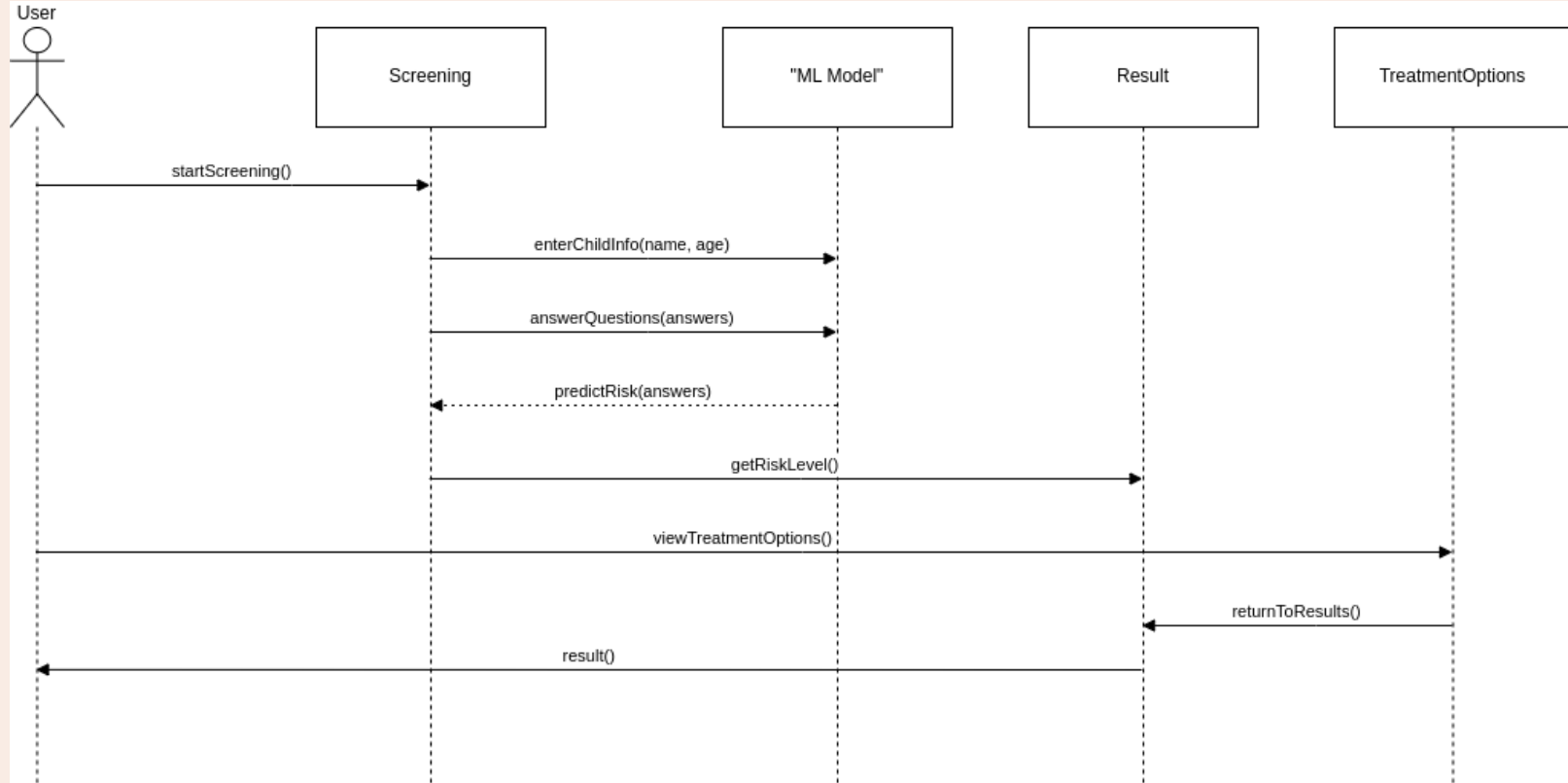
FLOW DIAGRAM



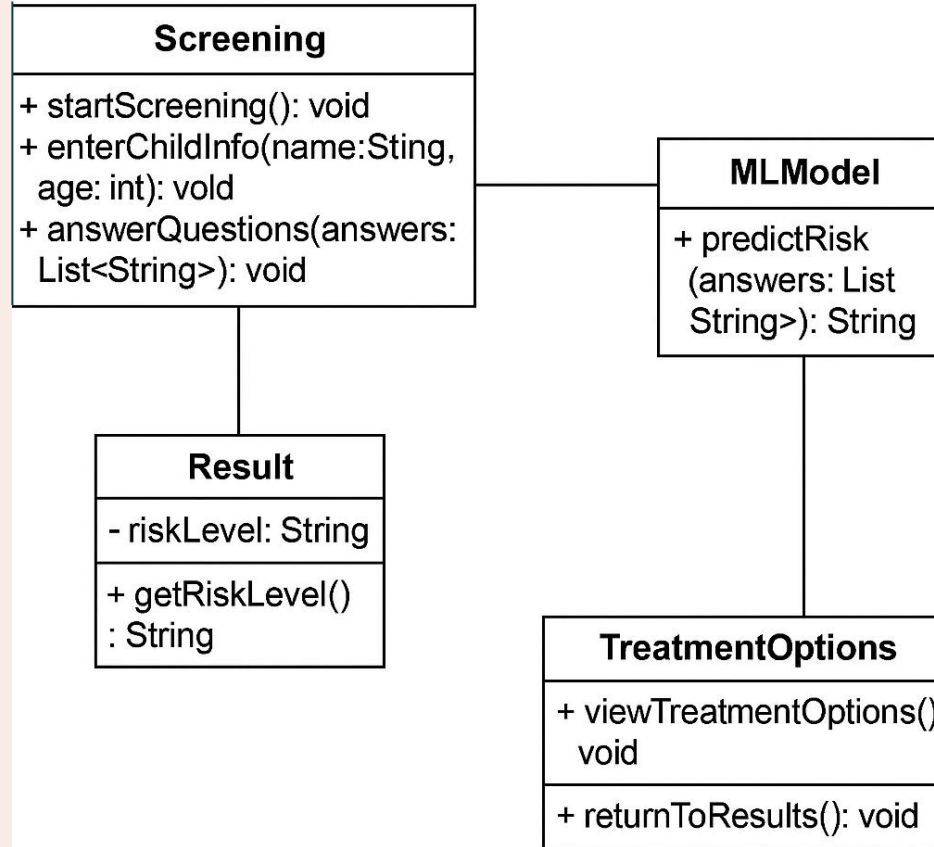
USE CASE DIAGRAM



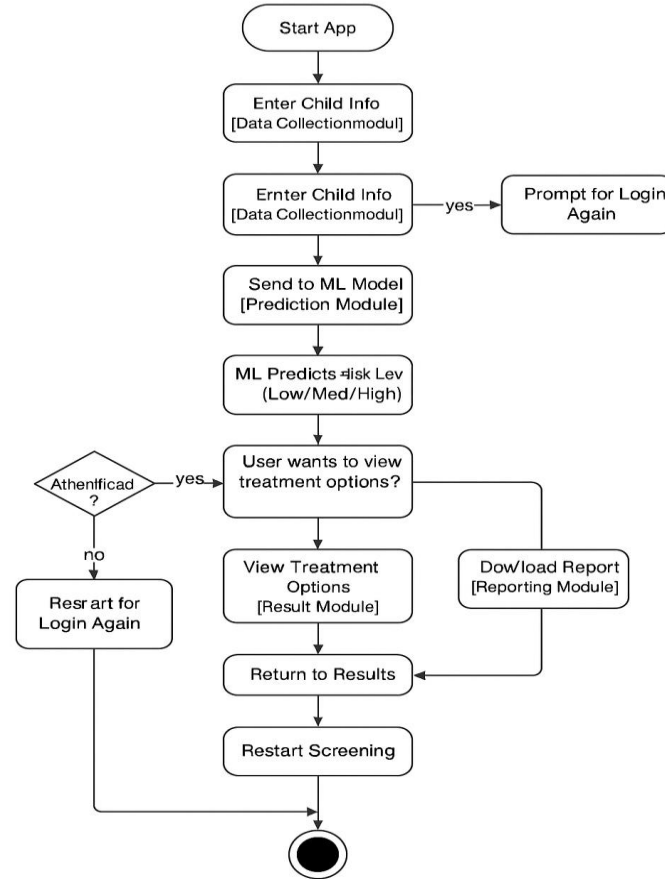
Sequence Diagram



CLASS DIAGRAM



ACTIVITY DIAGRAM



Sample Code for Training

Data Preprocessing

```
data = pd.read_csv("AutismDataset.csv") data.drop(columns=["Case_No", "Qchat-10-Score"], inplace=True)
```

Encode categorical columns

```
for col in data.select_dtypes(include=['object']):  
    data[col] = LabelEncoder().fit_transform(data[col])
```

Feature Selection and Splitting

```
X = data.drop(columns=["Class/ASD Traits "])  
y = data["Class/ASD Traits "] X_train, X_test, y_train, y_test = train_test_split(X, y,  
    stratify=y)
```

Model Training

```
model = XGBClassifier()  
model.fit(X_train, y_train) 5.5.1.4 Model Evaluation y_pred = model.predict(X_test)  
print(accuracy_score(y_test, y_pred))
```

Sample training code for front-end

Import Libraries and Load Model

```
from flask import Flask, render_template, request, redirect, url_for, jsonify  
import joblib, numpy as np
```

Load model and preprocessing tools

```
model = joblib.load("autism_model.pkl")  
scaler = joblib.load("scaler.pkl")  
label_encoders = joblib.load("label_encoders.pkl")
```

Initialize Flask App

```
app = Flask(__name__)  
details = {} # Temporary storage for user responses
```

Define Autism Survey Questions

```
questions = [ {"text": "Does your child look at you when you call his/her name?"},  
               {"text": "How easy is it for you to get eye contact with your child?"}, ... ]
```

Home Page Route

```
@app.route('/') def index(): return render_template('index.html')
```


Welcome Page

Autism Detection

[Home](#) [Features](#) [Contact](#) [FAQs](#) [About](#)

Autism Detection in Toddlers

Early detection helps in timely interventions that enhance development.

Start The Survey

Toddler Screen Input Screen

Autism Detection

[Home](#) [Features](#) [Contact](#) [FAQs](#) [About](#)

Child Screening Form

What is your child's age (in months)?

What is your child's sex?

☐ Male ☐ Female

What is your child's ethnicity?

Was your child born with jaundice?

☐ Yes ☐ No

Does your child have an immediate family member with Autism Spectrum Disorder (ASD)?

☐ Yes ☐ No

Who is completing this screening test?

Submit

Questionnaire

Autism Detection

[Home](#) [Features](#) [Contact](#) [FAQs](#) [About](#)

Question 1

Does your child look at you when you call his/her name?

Observe how your child responds when you call their name.

☐ Yes ☐ No

Next

Predictions results page

Autism Detection[Home](#)[Features](#)[Contact](#)[FAQs](#)[About](#)

Autism Screening Results

✓ Screening Result: **No**

Q-Chat Score

2

Risk Level

Low

Identified Risk Questions

Q1 Does your child look at you when you call his/her name?

Q2 How easy is it for you to get eye contact with your child?

Risk level display recommendations

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Recommended Treatment Options

✓ Low Risk Treatment Options

 **Parent Education and Training**

Learning strategies to support your child's development, communication, and behavior management at home.

[Learn more](#)

 **Speech and Language Therapy**

Helps develop communication skills, language comprehension, and social use of language.

[Learn more](#)

 **Developmental Monitoring**

Regular check-ups with pediatrician to track developmental milestones and address any concerns early.

[Learn more](#)

 **Social Engagement Activities**

Playgroups, storytime sessions, and other activities that promote social interaction and engagement.

[Learn more](#)

⚠ Medium Risk Treatment Options

⚠ High Risk Treatment Options

Test Cases

Test Case ID	Description	Input	Expected Output	Result
TC01	Valid Data	Complete and correctly formatted toddler details	Risk Level (Low / Medium / High)	Pass
TC02	Empty Field	Missing value in age field	Error Message prompting user to complete input	Pass
TC03	Invalid Format	Text entered in a numeric field (e.g., "Two" instead of 2)	Validation Error or form highlight	Pass
TC04	ML Accuracy	Pre-labeled test dataset entries	Model predicts correct labels with high accuracy	Pass
TC05	Load Test	50+ users submitting data simultaneously	Consistent server performance and response time	Pass
TC06	Model Availability	API call to prediction route while model is loading	Graceful fallback or "Model Loading" message	Pass
TC07	UI Feedback	After form submission	Result display with prediction and risk factors	Pass

Test Cases

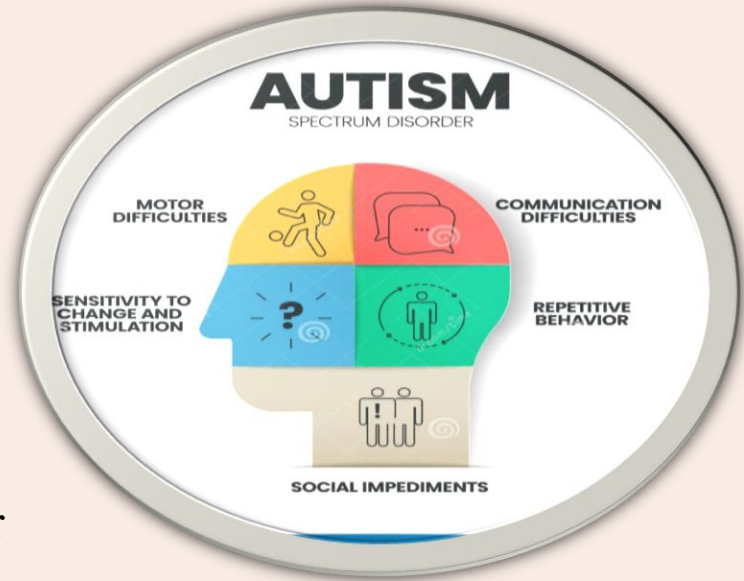
Test Case ID	Description	Input	Expected Output	Result
TC08	Risk Factor Breakdown	Valid input with mix of 0s and 1s for Q-Chat	List of at-risk questions and total score	Pass
TC09	Persistence Check	Predict, reload page, predict again	No crash or memory issue; consistent output	Pass
TC10	Invalid Route	GET on Predict data	Error on Delay	Pass
TC11	Security Test	Script injection	No execution	Pass
TC12	API Timeout	Slow Network	Fallback or fast	Fail
TC13	Incomplete form	Skip questions	Error Message	Fail
TC14	Corrupt Model	Bad model file	Error Message	Fail

Conclusion

Early detection of autism in toddlers is a collaborative effort between families and healthcare providers.

Accessing community resources and support groups can provide valuable assistance for families.

Continued research and awareness are vital for improving autism detection and intervention strategies.



Future Enhancements

- ❖ **Incorporate speech and facial expression analysis** for more comprehensive autism detection.
- ❖ Extend support to a **broader age range**, including older children and adolescents.
- ❖ **Develop a mobile app** version for wider accessibility and offline screening.
- ❖ Integrate with pediatric **Electronic Health Records (EHRs)** for seamless clinical use.
- ❖ Add an **AI-powered chatbot** to assist users with queries and guide them through the screening process.

References

UCI ASD Dataset: Source of real-world data for autism screening model training.

Q-CHAT-10 Research Papers:

- ❖ Allison, C., et al. (2012). *The Quantitative Checklist for Autism in Toddlers (Q-CHAT)*. Psychological Assessment.
- ❖ Baron-Cohen, S., et al. (2001). *The Checklist for Autism in Toddlers (CHAT)*. Journal of Autism and Developmental Disorders.

AI in Autism Research:

- ❖ Kosmicki, J. A., et al. (2015). *Searching for a minimal set of behaviors for autism detection through feature selection-based machine learning*. Translational Psychiatry.

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Machine Learning for Autism Detection:

- ❖ Thabtah, F. (2017). *Autism spectrum disorder screening: Machine learning adaptation and DSM-5 fulfillment*. Proceedings of the 1st International Conference on Medical and Health Informatics.
- ❖ Bone, D., et al. (2016). *Use of machine learning to improve autism screening and diagnostic tools: Effectiveness, challenges and future directions*. Journal of Autism and Developmental Disorders.

THANK YOU